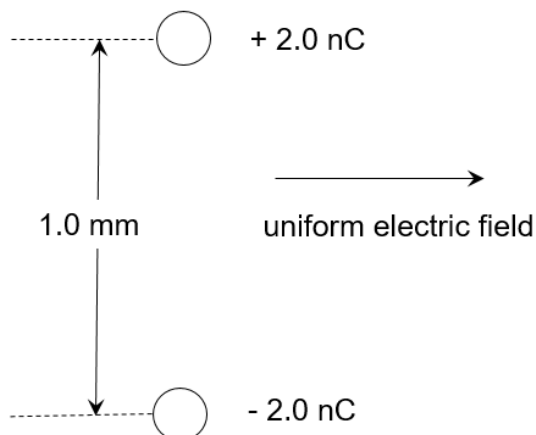


21 Two charges are fixed at a distance of 1.0 mm apart as shown below.

A uniform electric field with field strength 4.0 N C^{-1} is applied in the direction from left to right.

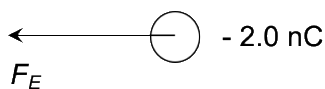
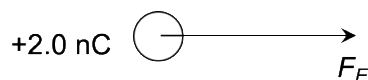


Which statement describes the net force and torque experienced by the pair of charges?

- | | |
|----------|--|
| A | No net force and no torque acts. |
| B | No net force acts and the torque is $8.0 \times 10^{-12} \text{ N m}$ in the clockwise direction. |
| C | A net force of $3.6 \times 10^{-14} \text{ N}$ acts and the torque is $4.0 \times 10^{-12} \text{ N m}$ in the clockwise direction |
| D | A net force of $3.6 \times 10^{-14} \text{ N}$ acts and the torque is $8.0 \times 10^{-12} \text{ N m}$ in the clockwise direction |

Answer: B

An electric dipole context where the two charges are connected by an electric force of attraction



$$\begin{aligned}
 \text{torque} &= qEd \\
 &= (2.0 \times 10^{-9})(4.0)(1.0 \times 10^{-3}) \\
 &= 8.0 \times 10^{-12} \text{ N m}
 \end{aligned}$$

22	A battery causes a current of 3 A to flow through a metal wire of diameter 2.0 mm. The number
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